Probability

ClipSAT Printable Practice Suite (clipsat.org) • Topic 11.01

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Cross sections of solids**

A survey was conducted on students' preferred modes of transportation to school. The results are summarized in the two-way frequency table below. \\ | | Bus | Bicycle | Walking | Total | |-----------------|-----|---------|---------|-------| | Under 16 | 45 | 30 | 25 | 100 | | 16 and Over | 60 | 15 | 25 | 100 | | Total | 105 | 45 | 50 | 200 | Let B be the event that a randomly selected student prefers to travel by Bus. Let U be the event that a randomly selected student is Under 16 years old. \\ What is the probability that a student prefers to travel by Bus, given that they are Under 16 years old? Mathematically, this is denoted as $P(B|U)$.

We are asked to find the conditional probability $P(B|U)$, which represents the probability that a student prefers to travel by Bus given that they are Under 16 years old. \\ The formula for conditional probability is $P(B|U) = \frac{P(B \cap U)}{P(U)}$. \\ From the table: - The number of students who prefer Bus AND are Under 16 ($B \cap U$) is 45. \\ - The total number of students who are Under 16 (U) is 100. \\ Therefore, $P(B \cap U) = \frac{45}{200}$ and $P(U) = \frac{100}{200}$. \\ Substituting these values.

2 **[Foundational]** • **Target: Solids of revolution**

A civil engineer measures the width of a river BC by setting up parallel stakes such that $DE \parallel BC$. Point D is on line AB and E is on line AC. If $AD = 45$ m, $DB = 75$ m, and $DE = 36$ m, what is the width of the river BC?

First compute the total line length: $AB = AD + DB = 45 + 75 = 120$ m. Because $DE \parallel BC$, $\Delta ADE \sim \Delta ABC$. The proportion of corresponding sides is $\frac{DE}{BC} = \frac{AD}{AB} \Rightarrow \frac{36}{BC} = \frac{45}{120} = \frac{3}{8}$. Cross-multiplying: $3 \cdot BC = 36 \times 8 = 288 \Rightarrow BC = 96$ m.

3 **[Foundational]** • **Target: Volume of prisms and cylinders**

A 5.5-foot-tall surveyor casts a 4.4-foot shadow on level ground. At the exact same moment, a nearby radio antenna tower casts a 36-foot shadow. How tall is the antenna tower?

Sunlight rays strike level ground at the same angle of elevation, creating similar right triangles between objects and their shadows: $\frac{\text{Height}}{\text{Shadow}} = \frac{5.5}{4.4} = 1.25$. For the tower: $\frac{H}{36} = 1.25 \Rightarrow H = 36 \times 1.25 = 45$ ft.

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4 [Foundational] • Target: Volume of pyramids and cones

In ΔABC, segment $DE \parallel BC$. The segment lengths are $AD = x$, $DB = x + 3$, $AE = x + 2$, and $EC = x + 7$. What is the value of x?

By the Triangle Proportionality Theorem, $\frac{AD}{DB} = \frac{AE}{EC} \Rightarrow \frac{x}{x+3} = \frac{x+2}{x+7}$. Cross-multiplying: $x(x + 7) = (x + 3)(x + 2) \Rightarrow x^2 + 7x = x^2 + 5x + 6$. Subtracting x^2 from both sides gives $7x = 5x + 6 \Rightarrow 2x = 6 \Rightarrow x = 3$.

5 [Foundational] • Target: Volume of spheres and hemispheres

In ΔABC, ray AD bisects $\angle BAC$ with D lying on side BC. If $AB = x + 3$, $AC = 2x$, $BD = 4$, and $DC = 6$, what is the value of x?

By the Angle Bisector Theorem, the ratio of the divided base segments equals the ratio of the adjacent sides: $\frac{BD}{DC} = \frac{AB}{AC} \Rightarrow \frac{4}{6} = \frac{x+3}{2x}$. Cross-multiplying: $4(2x) = 6(x + 3) \Rightarrow 8x = 6x + 18 \Rightarrow 2x = 18 \Rightarrow x = 9$.

6 [Foundational] • Target: Volume of composite solids

In an hourglass figure where $AB \parallel CD$ with intersection at E, $AE = 2x + 1$, $EC = 5x - 4$, $BE = 6$, and $ED = 9$. What is the value of x?

By similarity $\Delta ABE \sim \Delta CDE$: $\frac{AE}{EC} = \frac{BE}{ED} \Rightarrow \frac{2x+1}{5x-4} = \frac{6}{9} = \frac{2}{3}$. Cross-multiplying:

$3(2x + 1) = 2(5x - 4) \Rightarrow 6x + 3 = 10x - 8 \Rightarrow 4x = 11 \Rightarrow x = \frac{11}{4} = 2.75$.

7 [Foundational] • Target: Cross sections of solids

In right ΔABC with altitude $CD \perp AB$, $AD = 2$ and $DB = 8$. What is the ratio of the legs $\frac{AC}{BC}$?

By the Geometric Mean Leg Theorems: $AC^2 = AB \cdot AD$ and $BC^2 = AB \cdot DB$. Dividing the two equations: $\frac{AC^2}{BC^2} = \frac{AD}{DB} = \frac{2}{8} = \frac{1}{4}$. Taking the square root gives $\frac{AC}{BC} = \sqrt{\frac{1}{4}} = \frac{1}{2}$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Solids of revolution

Two similar triangles have areas of $45\ \text{cm}^2$ and $80\ \text{cm}^2$. If the perimeter of the smaller triangle is $27\ \text{cm}$, what is the perimeter of the larger triangle?

The ratio of the areas is $\frac{A_2}{A_1} = \frac{80}{45} = \frac{16}{9} = k^2$. The linear scale factor is $k = \sqrt{\frac{16}{9}} = \frac{4}{3}$. Perimeters scale linearly by k: $P_2 = 27 \times \frac{4}{3} = 36\ \text{cm}$.

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9 [Intermediate] • Target: Volume of prisms and cylinders

The point P has coordinates $(-3, 4)$. Point P' is the image of P after a reflection across the line $y = x$. What are the coordinates of P'?

When a point (x, y) is reflected across the line $y = x$, its image has coordinates (y, x). Given the point P has coordinates $(-3, 4)$. Applying the reflection rule, the coordinates of P' will be $(4, -3)$.

10 [Intermediate] • Target: Volume of pyramids and cones

In right ΔABC with right angle at C and altitude $CD \perp AB$, leg $BC = 12$ and hypotenuse $AB = 16$. What is the length of segment DB?

By the Geometric Mean Leg Theorem, $BC^2 = AB \cdot DB$. Substituting the known lengths: $12^2 = 16 \cdot DB \Rightarrow 144 = 16 \cdot DB \Rightarrow DB = \frac{144}{16} = 9$. (Consequently, $AD = 16 - 9 = 7$).

11 [Intermediate] • Target: Volume of prisms and cylinders

For the model $f(x) = 5x + 8$ in volume of prisms and cylinders, evaluate $f(5)$.

Final Answer: 33. Substitute $x = 5$: $f(5) = 5(5) + 8 = 33$.

12 [Intermediate] • Target: Volume of prisms and cylinders

Find the positive solutions to the quadratic equation $x^2 - 10x + 24 = 0$.

Final Answer: $x = 4$ and $x = 6$. Factoring $(x - 4)(x - 6) = 0$ yields the roots $x = 4$ and $x = 6$.

13 [Intermediate] • Target: Volume of pyramids and cones

For the model $f(x) = 6x + 9$ in volume of pyramids and cones, evaluate $f(2)$.

Final Answer: 21. Substitute $x = 2$: $f(2) = 6(2) + 9 = 21$.

14 [Intermediate] • Target: Volume of pyramids and cones

Find the positive solutions to the quadratic equation $x^2 - 5x + 4 = 0$.

Final Answer: $x = 1$ and $x = 4$. Factoring $(x - 1)(x - 4) = 0$ yields the roots $x = 1$ and $x = 4$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Volume of pyramids and cones • Cross-Lesson: 6.01 Polygons & Quadrilaterals

A data set recorded for volume of pyramids and cones consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 22.5. The sum is $16 + 20 + 24 + 30 = 90$, and dividing by 4 gives 22.5.

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16 [Challenging] • Target: Volume of pyramids and cones • Cross-Lesson: 7.01 Circles

For the model $f(x) = 6x + 9$ in volume of pyramids and cones, evaluate $f(3)$.

Final Answer: 27. Substitute $x = 3$: $f(3) = 6(3) + 9 = 27$.

17 [Challenging] • Target: Volume of pyramids and cones • Cross-Lesson: 8.01 Area, Surface Area & Volume

Find the positive solutions to the quadratic equation $x^2 - 7x + 10 = 0$.

Final Answer: $x = 2$ and $x = 5$. Factoring $(x - 2)(x - 5) = 0$ yields the roots $x = 2$ and $x = 5$.

18 [Challenging] • Target: Volume of pyramids and cones • Cross-Lesson: 9.01 Transformations & Symmetry

For the model $f(x) = 6x + 9$ in volume of pyramids and cones, evaluate $f(4)$.

Final Answer: 33. Substitute $x = 4$: $f(4) = 6(4) + 9 = 33$.

19 [Challenging] • Target: Volume of pyramids and cones • Cross-Lesson: 1.01 Foundations & Reasoning

Find the positive solutions to the quadratic equation $x^2 - 9x + 18 = 0$.

Final Answer: $x = 3$ and $x = 6$. Factoring $(x - 3)(x - 6) = 0$ yields the roots $x = 3$ and $x = 6$.

20 [Challenging] • Target: Volume of pyramids and cones • Cross-Lesson: 10.01 Coordinate Geometry

For the model $f(x) = 6x + 9$ in volume of pyramids and cones, evaluate $f(5)$.

Final Answer: 39. Substitute $x = 5$: $f(5) = 6(5) + 9 = 39$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Cross sections of solids

For the model $f(x) = 3x + 6$ in cross sections of solids, evaluate $f(2)$.

a. 12 ✓

b. 15

c. 6

d. 16

Substitute $x = 2$: $f(2) = 3(2) + 6 = 12$.

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22 [Foundational] • Target: Cross sections of solids

Find the positive solutions to the quadratic equation $x^2 - 8x + 12 = 0$.

a. $x = 2$ and $x = 7$

b. $x = 2$ **and** $x = 6$ ✓

c. $x = -2$ and $x = -6$

d. $x = 8$ and $x = 12$

Factoring $(x - 2)(x - 6) = 0$ yields the roots $x = 2$ and $x = 6$.

23 [Foundational] • Target: Cross sections of solids

For the model $f(x) = 3x + 6$ in cross sections of solids, evaluate $f(5)$.

a. 21 ✓

b. 24

c. 15

d. 25

Substitute $x = 5$: $f(5) = 3(5) + 6 = 21$.

24 [Foundational] • Target: Solids of revolution

For the model $f(x) = 4x + 7$ in solids of revolution, evaluate $f(2)$.

a. 15 ✓

b. 19

c. 8

d. 19

Substitute $x = 2$: $f(2) = 4(2) + 7 = 15$.

25 [Foundational] • Target: Solids of revolution

Find the positive solutions to the quadratic equation $x^2 - 9x + 14 = 0$.

a. $x = 2$ and $x = 8$

b. $x = 2$ **and** $x = 7$ ✓

c. $x = -2$ and $x = -7$

d. $x = 9$ and $x = 14$

Factoring $(x - 2)(x - 7) = 0$ yields the roots $x = 2$ and $x = 7$.

26 [Foundational] • Target: Solids of revolution

For the model $f(x) = 4x + 7$ in solids of revolution, evaluate $f(5)$.

a. 27 ✓

b. 31

c. 20

d. 31

Substitute $x = 5$: $f(5) = 4(5) + 7 = 27$.

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27 [Foundational] • Target: Volume of prisms and cylinders

For the model $f(x) = 5x + 8$ in volume of prisms and cylinders, evaluate $f(2)$.

a. 18 ✓

b. 23

c. 10

d. 22

Substitute $x = 2$: $f(2) = 5(2) + 8 = 18$.

28 [Foundational] • Target: Volume of prisms and cylinders

Find the positive solutions to the quadratic equation $x^2 - 6x + 8 = 0$.

a. $x = 2$ and $x = 5$

b. $x = 2$ **and** $x = 4$ ✓

c. $x = -2$ and $x = -4$

d. $x = 6$ and $x = 8$

Factoring $(x - 2)(x - 4) = 0$ yields the roots $x = 2$ and $x = 4$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Cross sections of solids

Find the positive solutions to the quadratic equation $x^2 - 6x + 5 = 0$.

a. $x = 1$ and $x = 6$

b. $x = 1$ **and** $x = 5$ ✓

c. $x = -1$ and $x = -5$

d. $x = 6$ and $x = 5$

Factoring $(x - 1)(x - 5) = 0$ yields the roots $x = 1$ and $x = 5$.

30 [Intermediate] • Target: Cross sections of solids

A data set recorded for cross sections of solids consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

a. 17.5

b. 18.5

c. 19.5 ✓

d. 21.5

The sum is $13 + 17 + 21 + 27 = 78$, and dividing by 4 gives 19.5.

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31 [Intermediate] • Target: Cross sections of solids

For the model $f(x) = 3x + 6$ in cross sections of solids, evaluate $f(3)$.

a. 15 ✓

b. 18

c. 9

d. 19

Substitute $x = 3$: $f(3) = 3(3) + 6 = 15$.

32 [Intermediate] • Target: Cross sections of solids

Find the positive solutions to the quadratic equation $x^2 - 10x + 21 = 0$.

a. $x = 3$ and $x = 8$

b. $x = 3$ **and** $x = 7$ ✓

c. $x = -3$ and $x = -7$

d. $x = 10$ and $x = 21$

Factoring $(x - 3)(x - 7) = 0$ yields the roots $x = 3$ and $x = 7$.

33 [Intermediate] • Target: Cross sections of solids

Find the positive solutions to the quadratic equation $x^2 - 12x + 32 = 0$.

a. $x = 4$ and $x = 9$

b. $x = 4$ **and** $x = 8$ ✓

c. $x = -4$ and $x = -8$

d. $x = 12$ and $x = 32$

Factoring $(x - 4)(x - 8) = 0$ yields the roots $x = 4$ and $x = 8$.

34 [Intermediate] • Target: Solids of revolution

A data set recorded for solids of revolution consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

a. 18.5

b. 19.5

c. 20.5 ✓

d. 22.5

The sum is $14 + 18 + 22 + 28 = 82$, and dividing by 4 gives 20.5.

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35 [Intermediate] • Target: Solids of revolution

For the model $f(x) = 4x + 7$ in solids of revolution, evaluate $f(3)$.

a. 19 ✓

b. 23

c. 12

d. 23

Substitute $x = 3$: $f(3) = 4(3) + 7 = 19$.

36 [Intermediate] • Target: Solids of revolution

Find the positive solutions to the quadratic equation $x^2 - 11x + 24 = 0$.

a. $x = 3$ and $x = 9$

b. $x = 3$ **and** $x = 8$ ✓

c. $x = -3$ and $x = -8$

d. $x = 11$ and $x = 24$

Factoring $(x - 3)(x - 8) = 0$ yields the roots $x = 3$ and $x = 8$.

37 [Intermediate] • Target: Solids of revolution

Find the positive solutions to the quadratic equation $x^2 - 13x + 36 = 0$.

a. $x = 4$ and $x = 10$

b. $x = 4$ **and** $x = 9$ ✓

c. $x = -4$ and $x = -9$

d. $x = 13$ and $x = 36$

Factoring $(x - 4)(x - 9) = 0$ yields the roots $x = 4$ and $x = 9$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Volume of prisms and cylinders • Cross-Lesson: 9.01 Transformations & Symmetry

A data set recorded for volume of prisms and cylinders consists of the four values $\{15, 19, 23, 29\}$.

What is the arithmetic mean?

a. 19.5

b. 20.5

c. 21.5 ✓

d. 23.5

The sum is $15 + 19 + 23 + 29 = 86$, and dividing by 4 gives 21.5.

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39 [Challenging] • Target: Volume of prisms and cylinders • Cross-Lesson: 1.01 Foundations & Reasoning

For the model $f(x) = 5x + 8$ in volume of prisms and cylinders, evaluate $f(3)$.

a. 23 ✓

b. 28

c. 15

d. 27

Substitute $x = 3$: $f(3) = 5(3) + 8 = 23$.

40 [Challenging] • Target: Volume of prisms and cylinders • Cross-Lesson: 10.01 Coordinate Geometry

Find the positive solutions to the quadratic equation $x^2 - 8x + 15 = 0$.

a. $x = 3$ and $x = 6$

b. $x = 3$ **and** $x = 5$ ✓

c. $x = -3$ and $x = -5$

d. $x = 8$ and $x = 15$

Factoring $(x - 3)(x - 5) = 0$ yields the roots $x = 3$ and $x = 5$.

41 [Challenging] • Target: Volume of pyramids and cones • Cross-Lesson: 2.01 Lines & Angles

A data set recorded for volume of pyramids and cones consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean?

a. 20.5

b. 21.5

c. 22.5 ✓

d. 24.5

The sum is $16 + 20 + 24 + 30 = 90$, and dividing by 4 gives 22.5.

42 [Challenging] • Target: Volume of spheres and hemispheres • Cross-Lesson: 3.01 Triangles & Congruence

A data set recorded for volume of spheres and hemispheres consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean?

a. 21.5

b. 22.5

c. 23.5 ✓

d. 25.5

The sum is $17 + 21 + 25 + 31 = 94$, and dividing by 4 gives 23.5.

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43 [Challenging] • Target: Volume of spheres and hemispheres • Cross-Lesson: 4.01 Similarity & Proportional Reasoning

For the model $f(x) = 2x + 3$ in volume of spheres and hemispheres, evaluate $f(3)$.

a. 9 ✓

b. 11

c. 6

d. 13

Substitute $x = 3$: $f(3) = 2(3) + 3 = 9$.

44 [Challenging] • Target: Volume of composite solids • Cross-Lesson: 5.01 Right Triangles & Trigonometry

A data set recorded for volume of composite solids consists of the four values $\{18, 22, 26, 32\}$. What is the arithmetic mean?

a. 22.5

b. 23.5

c. 24.5 ✓

d. 26.5

The sum is $18 + 22 + 26 + 32 = 98$, and dividing by 4 gives 24.5.

45 [Challenging] • Target: Volume of composite solids • Cross-Lesson: 6.01 Polygons & Quadrilaterals

For the model $f(x) = 3x + 4$ in volume of composite solids, evaluate $f(3)$.

a. 13 ✓

b. 16

c. 9

d. 17

Substitute $x = 3$: $f(3) = 3(3) + 4 = 13$.

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